Exclusions using the marked generators of a regular cover

Human Compression project

We acquire the normalizer description of chiral quotients in Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Theorem 3.1. We keep every possible flag stabilizer, including those with nontrivial core. The cover action is not assumed faithful. Enumeration of normal subgroups remains an explicit software premise. We also use a necessary condition already implicit in the reconstruction of faces from a flag graph.

Proposition 1. *Let a regular polytope with group $W = \langle r_0, \dots, r_{d-1} \rangle$ cover a chiral abstract polytope, and let N stabilize its base flag. Put*

$$H = \langle r_0 r_1, \dots, r_0 r_{d-1} \rangle, \qquad W_i = \langle r_j : j \neq i \rangle.$$

Then $[W : H] = 2$, $N_W(N) = H$, and

$$N \trianglelefteq H, \qquad N \neq 1, \qquad N \not\trianglelefteq W.$$

The kernel of the cover action is $N \cap N^{r_0}$, which may be nontrivial. Moreover $\bigcap_{i=0}^{d-1} NW_i = N$.

Proof. The subgroup generated by the displayed pairs contains every product of two generators, since

$$r_i r_j = (r_0 r_i)^{-1} (r_0 r_j), \qquad r_0 r_0 = 1.$$

It consists of the even words and has index at most two. In a chiral flag graph every adjacency interchanges the two automorphism orbits. An odd word cannot stabilize a flag. The even subgroup therefore has index two and agrees with the normalizer of N, by the acquired normalizer theorem.

The normal-subgroup conditions follow from the same theorem. Since N is normal in the subgroup H of index two, its only conjugates in W are N, N^{r_0}. Their intersection is therefore the core and the kernel of the coset action. If $N = 1$, its normalizer would be all of W, contrary to chirality.

Identify graph vertices with right cosets Nw, with colour j sending Nw to Nwr_j. The face of rank i through the base flag is precisely the connected component obtained by omitting colour i. Its vertices are the cosets Nh with $h \in W_i$. Hence Nw has every base face if and only if $w \in NW_i$ for every i. A flag is determined by its faces; such a vertex must equal N, so $w \in N$. The reverse inclusion is immediate. □

For each literal source marking, the reader checks the generated group, its order, the involutions, adjacent orders and string commutations. The database identifier is a source label; no abstract identification is used by this test. If $H = W$, this marking is excluded. Otherwise it enumerates all normal subgroups of H and retains exactly the candidates satisfying the displayed group conditions. For each it computes all NW_i. If T_i is a right transversal of $N \cap W_i$ in W_i, then $NW_i = \coprod_{t \in T_i} Nt$; this is the finite computation used. A negative result also retains a literal element $w \notin N$ and checks $n_i^{-1} w \in W_i$ for some $n_i \in N$ at every rank.

Passing this test does not prove that the quotient is a polytope. Survivors still require the full face and flag axioms and comparison up to coloured graph isomorphism. No new exclusion is credited before the individual source entry has finished and been reviewed. The earlier filter for faithful actions alone does not exclude arbitrary cover actions. The source examples and original normalizer enumeration retain their original attribution. This component supplies an explicit finite application of the acquired group and face criteria, not a new complete atlas theorem.